INCH-POUND

MIL-M-28787/414
23 MARCH 1994

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-1397 TYPE E SERIAL INTERFACE ADAPTER MODULE,

KEY CODE ATC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic MIL-STD-1397 type E serial interface adapter module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.222 failures per million operating hours for class I or 2.446 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be ATC.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I Module electrical requirements

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	S	1	INF	1	INF	1	INF	Mohm
A2-A50	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	X	1	INF	1	INF	1	INF	Mohm
B1-B8	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Power supply stress	IV	S	----------Module shall pass all remaining tests----------						
D1	Ibd	V	S	-0 02	1 02	-0 02	1 02	-0 02	1.02	mA
D2-D6	Ibd	V	S	0 0	0 1	0 0	0 1	0.0	0.1	mA
D7	Ibd	V	S	-0 58	1 14	-0 58	1 14	-0.58	1.14	mA
D8	Ibd	V	S	0 0	0.1	0 0	0.1	0.0	0.1	mA
D9	Ibd	V	S	-0 02	0 22	-0 02	0 22	-0.02	0.22	mA
D10,D11	Ibd	V	S	-0.02	1 02	-0 02	1 02	-0.02	1.02	mA
D12	Ibd	V	S	0 0	0 1	0 0	0 1	0.0	0.1	mA
D13	Ibd	V	S	-0 02	1 02	-0 02	1.02	-0.02	1.02	mA
D14	Ibd	V	S	0 0	0 1	0 0	0 1	0.0	0.1	mA
E1-E12	Ios	VI	S	-100.0	-40 0	-100 0	-40.0	-100.0	-40.0	mA
E13	Ios	VI	S	-85.0	-15 0	-85 0	-15 0	-85 0	-15.0	mA
E14	Ios	VI	S	-100.0	-40 0	-100 0	-40 0	-100 0	-40.0	mA
E15	Ios	VI	S	-85.0	-15.0	-85 0	-15.0	-85.0	-15 0	mA
E16	Ios	VI	S	-100.0	-40 0	-100 0	-40 0	-100 0	-40 0	mA
F1	Iih	VII	X	-3 04	-0.5	-3.04	-0.5	-3.04	-0.5	mA
F2-F6	Iih	VII	X	-0.4	0 02	-0 4	0.02	-0.4	0.02	mA
F7	Iih	VII	X	-6 64	-0 35	-6 64	-0.35	-6.64	-0.35	mA
F8	Iih	VII	X	-0.4	0.02	-0 4	0.02	-0.4	0.02	mA
F9	Iih	VII	X	-1.64	-0.51	-1 64	-0.51	-1.64	-0.51	mA
F10,F11	Iih	VII	X	-3 04	-0.5	-3 04	-0.5	-3.04	-0.5	mA
F12	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F13	Iih	VII	X	-3.04	-0.5	-3.04	-0.5	-3.04	-0.5	mA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F14	Ilh	VII	X	-0 4	0 02	-0 4	0 02	-0 4	0 02	mA
G1	III	VIII	X	-3 04	-0.5	-3 04	-0.5	-3.04	-0.5	mA
G2-G6	III	VIII	X	-0 4	0 02	-0 4	0 02	-0.4	0 02	mA
G7	III	VIII	X	-6 64	-0 35	-6 64	-0 35	-6 64	-0.35	mA
G8	III	VIII	X	-0 4	0.02	-0 4	0 02	-0.4	0 02	mA
G9	III	VIII	X	-1.64	-0 51	-1 64	-0 51	-1 64	-0 51	mA
G10,G11	III	VIII	X	-3 04	-0 5	-3 04	-0 5	-3.04	-0 5	mA
G12	III	VIII	X	-0.4	0 02	-0 4	0 02	-0.4	0.02	mA
G13	III	VIII	X	-3.04	-0 5	-3.04	-0.5	-3.04	-0.5	mA
G14	III	VIII	X	-0.4	0 02	-0 4	0.02	-0.4	0.02	mA
H1-H16	Voh	IX	X	2.4	4 5	2 4	4.5	2.4	4.5	V
J1-J16	Vol	X	X	0.0	0.5	0.0	0 5	0.0	0.5	V
L1-L8	Vic	XI	S	-1.5	0.0	-1 5	0 0	-1.5	0.0	V
M1	PSFC	XII	S	2.3	4.3	2.3	4.3	2.3	4.3	µF
M2	PSFC	XII	S	4.5	8 3	4 5	8 3	4 5	8 3	µF
M3	PSFC	XII	S	2.4	4.6	2 4	4.6	2.4	4 8	µF
N1	Icc	XIII	X	520	1230	520	1230	520	1230	mA
N2	Icc	XIII	X	-249	-100	-249	-100	-249	-100	mA
P1,P2	Functional	XIV	X	See table XIV						
Q1,Q2	Dynamic	XV	S	See table XV						
Q3,Q4	Dynamic	XV	Q	See table XV						
R1-R6	Cin	XVII	Q	2	75	2	75	2	75	pF
R7	Cin	XVII	Q	10	235	10	235	10	235	pF
R8	Cin	XVII	Q	2	100	2	100	2	100	pF
R9	Cin	XVII	Q	2	125	2	125	2	125	pF
R10	Cin	XVII	Q	4	75	4	75	4	75	pF
R11	Cin	XVII	Q	4	100	4	100	4	100	pF

-See footnotes at end of table

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R12	Cin	XVII	Q	2	75	2	75	2	75	pF
R13	Cin	XVII	Q	4	75	4	75	4	75	pF
R14	Cin	XVII	Q	2	125	2	125	2	125	pF
T1,T2	PSTA	XVIII	Q	0	400	0	400	0	400	mV

- See footnotes on next page

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/414-1 for class I and M28787/414-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc and (V2) of -5.7 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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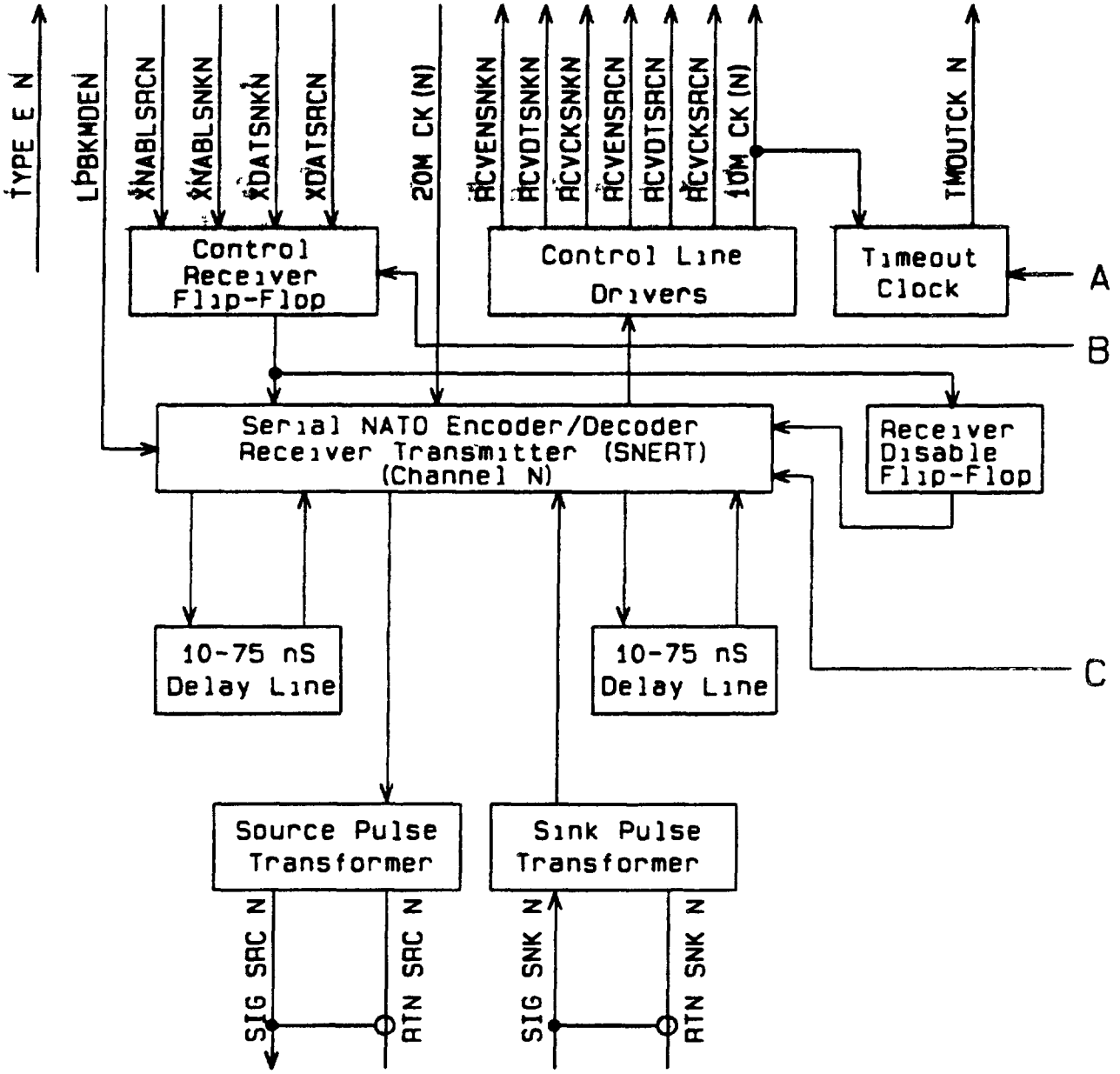


FIGURE 1. ATC functional block diagram.

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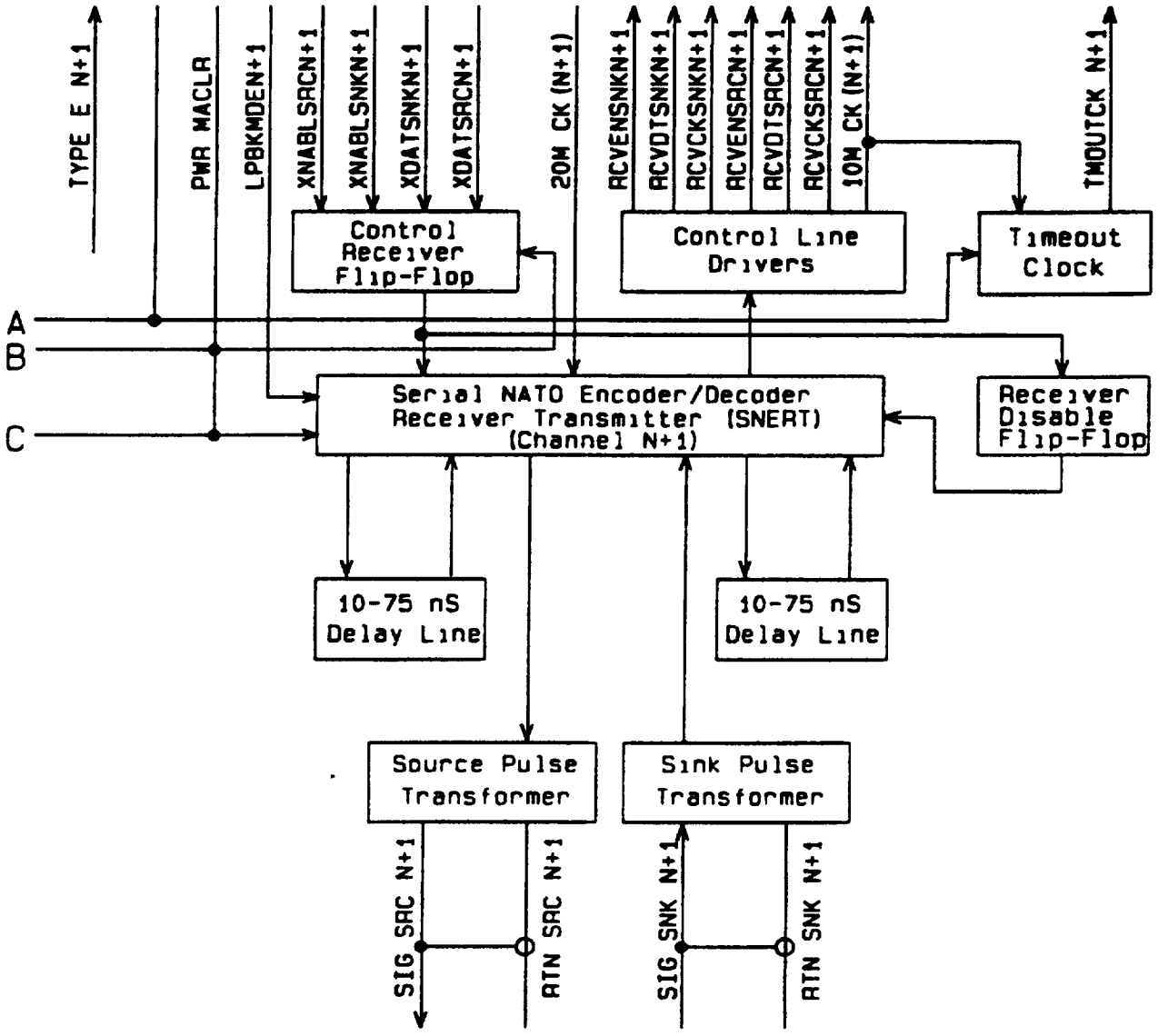


FIGURE 1. **ATC functional block diagram** - Continued.

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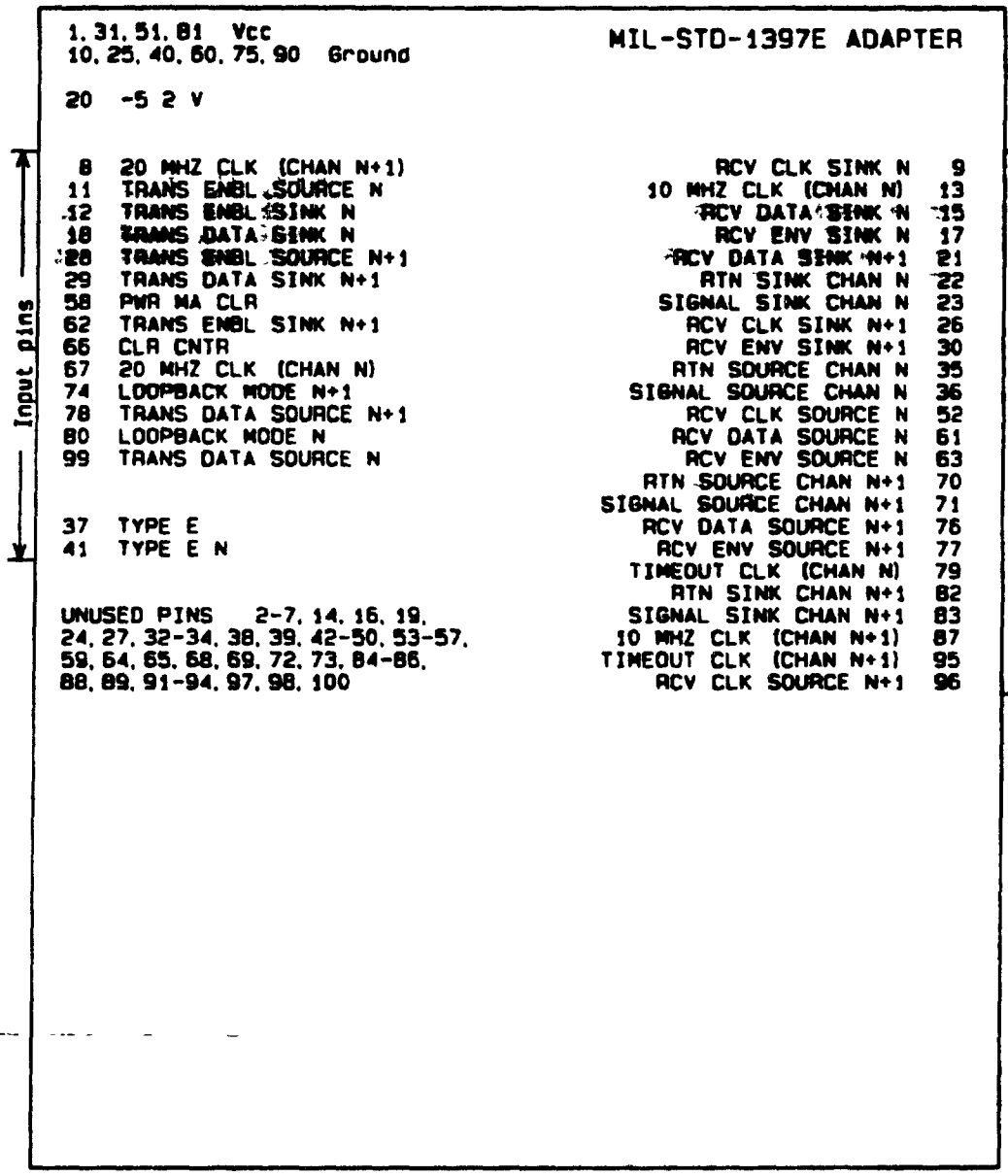


FIGURE 2. ATC pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

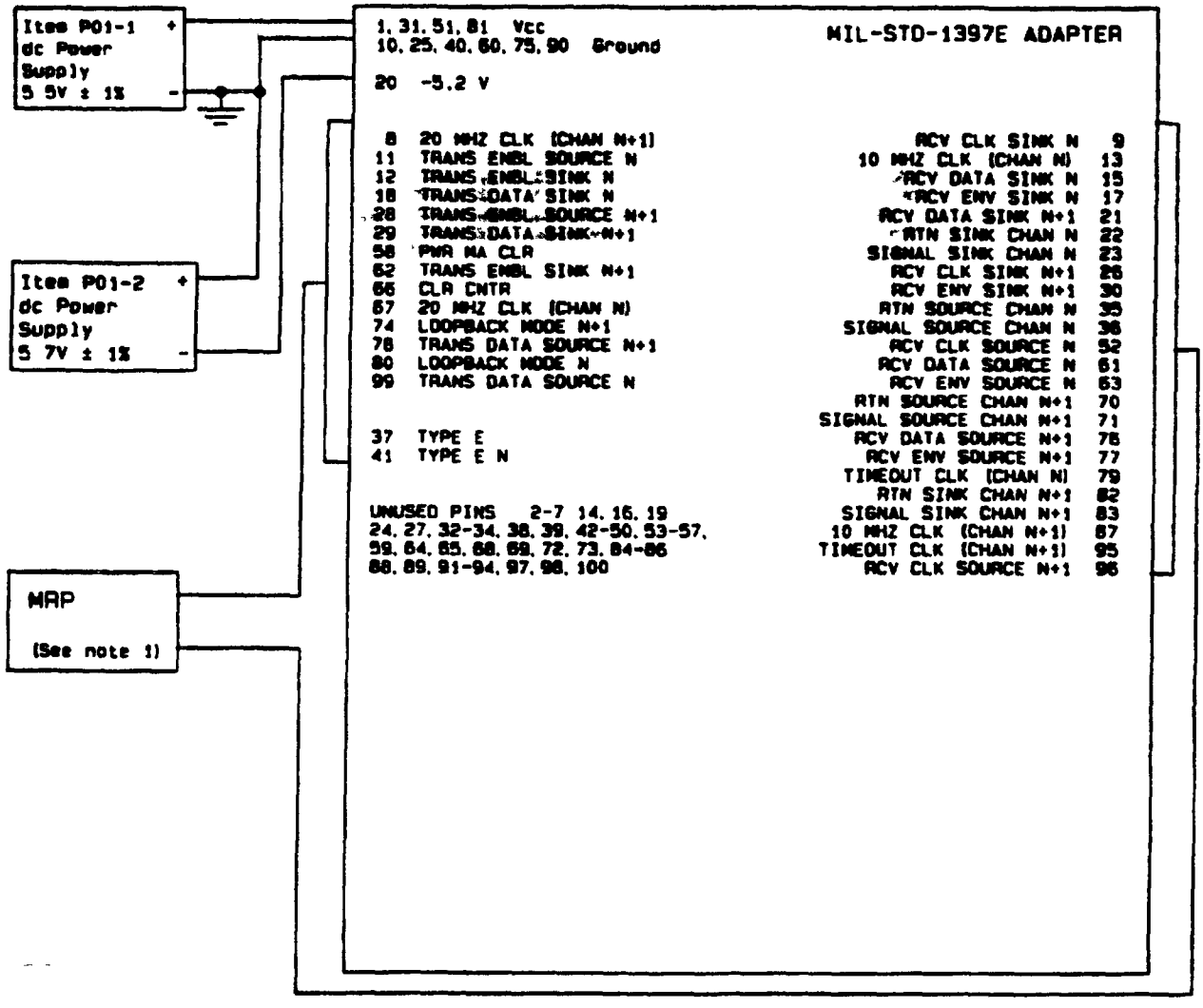
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. ATC operating temperature and component temperature test circuit.

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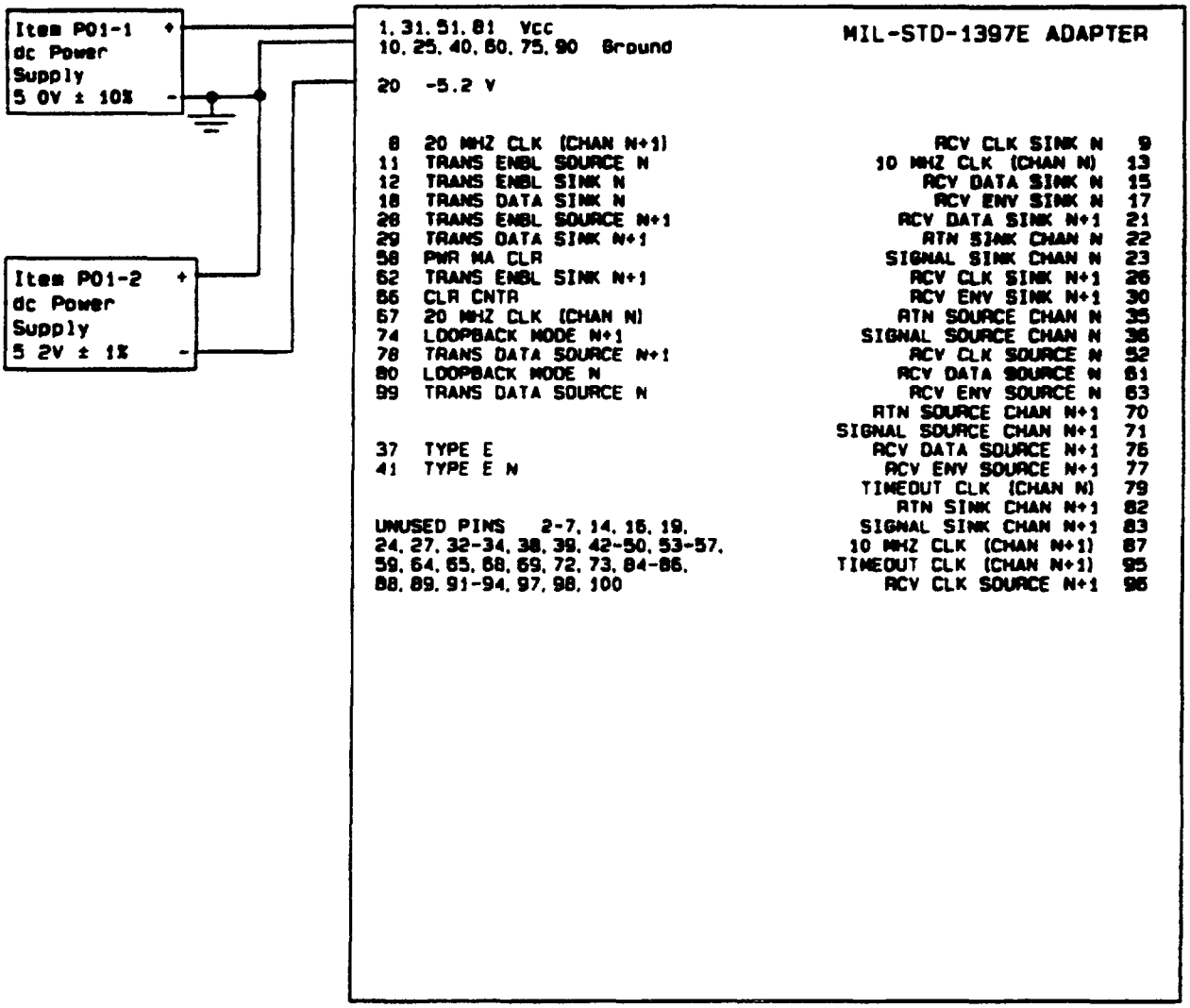


FIGURE 4. ATC vibration test circuit.

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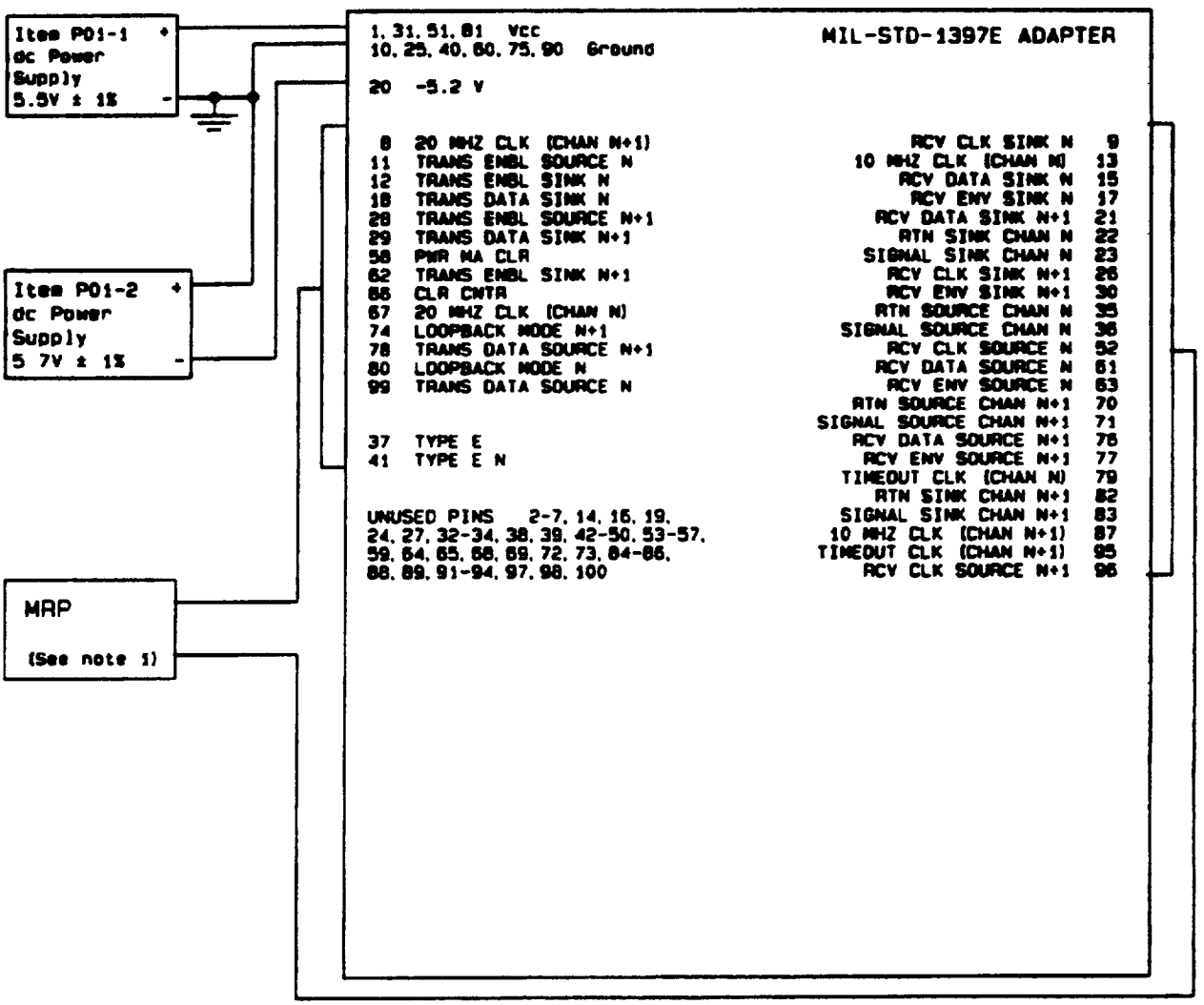
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. ATC life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Iin = input forcing current.
(g) Ioh = output high level current.
(h) Iol = output low level current.
(i) Ios = output short circuit current.
(j) NA = not applicable.
(k) PSFC = power supply filter capacitance.
(l) PSTA = power supply transient amplitude.
(m) Vic = input clamp diode voltage.
(n) Vih = input high level voltage.
(o) Vil = input low level voltage.
(p) Vin = input forcing voltage.
(q) Voh = output high level voltage.
(r) Vol = output low level voltage.
(s) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: (Vcc) 4.5 V dc to 5.5 V dc and (V2) -4.7 V dc to -5.7 V dc. Module power dissipation: 8.19 W (max); 5.49 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
20M CK(N+1)	NA	0.4	2.4	-3.04	-0.50	1.02	75
XNABLSRCN	-1.5	0.4	2.4	-0.4	0.02	0.1	75
XNABLSNKN	-1.5	0.4	2.4	-0.4	0.02	0.1	75
XDATSNKN	-1.5	0.4	2.4	-0.4	0.02	0.1	75
XNABLSRCN+1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
XDATSNKN+1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
XNABLSNKN+1	-1.5	0.4	2.4	-0.4	0.02	0.1	100
CLR CNTR	NA	0.4	2.4	-1.64	-0.51	0.22	125
20M CK(N)	NA	0.4	2.4	-3.04	-0.50	1.02	75
LPBKMDEN+1	NA	0.4	2.4	-3.04	-0.50	1.02	100
XDATSRCN+1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
LPBKMDEN	NA	0.4	2.4	-3.04	-0.50	1.02	75
XDATSRCN	-1.5	0.4	2.4	-0.4	0.02	0.1	125
PWR MACLR	NA	0.4	2.4	-6.64	-0.35	1.14	235

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)
RCVCKSNKN	2.4	0.5	-0.4	6.0	-40.0
10M CK(N)	2.4	0.5	-0.4	6.0	-40.0
RCVDTSNKN	2.4	0.5	-0.4	6.0	-40.0
RCVENSNKN	2.4	0.5	-0.4	6.0	-40.0
RCVDTSNKN+1	2.4	0.5	-0.4	6.0	-40.0
RCVCKSNKN+1	2.4	0.5	-0.4	6.0	-40.0
RCVENSNKN+1	2.4	0.5	-0.4	6.0	-40.0
RCVCKSRCN	2.4	0.5	-0.4	6.0	-40.0
RCVDTSRCN	2.4	0.5	-0.4	6.0	-40.0
RCVENSRCN	2.4	0.5	-0.4	6.0	-40.0
RCVDTSRCN+1	2.4	0.5	-0.4	6.0	-40.0
RCVENSRCN+1	2.4	0.5	-0.4	6.0	-40.0
TMOUTCK N	2.4	0.5	-0.16	1.6	-15.0

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)
10M CK(N+1)	2.4	0.5	-0.4	6.0	-40.0
TMOUTCK N+1	2.4	0.5	-0.16	1.6	-15.0
RCVCKSRCN+1	2.4	0.5	-0.4	6.0	-40.0
RTN SNK N	NA	NA	NA	NA	NA
SIG SNK N	2.4	0.5	NA	NA	NA
RTN SRC N	NA	NA	NA	NA	NA
SIG SRC N	2.4	0.5	NA	NA	NA
RTN SRC N+1	NA	NA	NA	NA	NA
SIG SRC N+1	2.4	0.5	NA	NA	NA
RTN SNK N+1	NA	NA	NA	NA	NA
SIG SNK N+1	NA	NA	NA	NA	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	RTN SRC N to RCVENSRCN	-	-	-	56	56	56	nS	12
		SIG SRC N to RCVENSRCN	-	-	-					
		RTN SRC N+1 to RCVENSRCN+1	-	-	-					
1		SIG SRC N+1 to RCVENSRCN+1	-	-	-	56	56	56		
2		RTN SRC N to RCVENSRCN	-	-	-	206	206	206		
		SIG SRC N to RCVENSRCN	-	-	-					
		RTN SRC N+1 to RCVENSRCN+1	-	-	-					
2		SIG SRC N+1 to RCVENSRCN+1	-	-	-	206	206	206		
3		RTN SRC N to RCVCKSRCN	-	-	-	176	176	176		
3	tpd	SIG SRC N to RCVCKSRCN	-	-	-	176	176	176	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	RTN SRC N+1 to RCVCKSRCN+1	-	-	-	176	176	176	nS	12
3		SIG SRC N+1 to RCVCKSRCN+1	-	-	-	176	176	176		
4		RTN SNK N to RCVENSNKN	-	-	-	56	56	56		
		SIG SNK N to RCVENSNKN	-	-	-					
		RTN SNK N+1 to RCVENSNKN+1	-	-	-					
4		SIG SNK N+1 to RCVENSNKN+1	-	-	-	56	56	56		
5		RTN SNK N to RCVENSNKN	-	-	-	206	206	206		
		SIG SNK N to RCVENSNKN	-	-	-					
		RTN SNK N+1 to RCVENSNKN+1	-	-	-					
5	tpd	SIG SNK N+1 to RCVENSNKN+1	-	-	-	206	206	206	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	RTN SNK N to RCVCKSNKN	-	-	-	176	176	176	nS	12
		SIG SNK N to RCVCKSNKN	-	-	-					
		RTN SNK N+1 to RCVCKSNKN+1	-	-	-					
6		SIG SNK N+1 to RCVCKSNKN+1	-	-	-	176	176	176		
7		20M CK(N) to 10M CK(N)	-	-	-	50	50	50		
7		20M CK(N+1) to 10M CK(N+1)	-	-	-	50	50	50		
8		20M CK(N) to 10M CK(N)	-	-	-	59	59	59		
8		20M CK(N+1) to 10M CK(N+1)	-	-	-	59	59	59		
9		20M CK(N) to TMOUTCK N	-	-	-	84	84	84		
9		20M CK(N+1) to TMOUTCK N+1	-	-	-					
10	tpd	20M CK(N) to TMOUTCK N	-	-	-	84	84	84	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tpd	20M CK(N+1) to TMOUTCK N+1	-	-	-	84	84	84	nS	12
11	tsu	XNABLSRCN to 10M CK(N)	12	12	12	-	-	-		
		XDATSRCN to 10M CK(N)				-	-	-		
		XNABLSNKN to 10M CK(N)				-	-	-		
		XDATSNKN to 10M CK(N)				-	-	-		
		XNABLSRCN+1 to 10M CK(N+1)				-	-	-		
		XDATSRCN+1 to 10M CK(N+1)				-	-	-		
		XNABLSNKN+1 to 10M CK(N+1)				-	-	-		
11	tsu	XDATSNKN+1 to 10M CK(N+1)	12	12	12	-	-	-		
12	th	10M CK(N) to XNABLSRCN	10	10	10	-	-	-		
12	th	10M CK(N) to XDATSRCN	10	10	10	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	th	10M CK(N) to XNABLSNKN	10	10	10	-	-	-	nS	12
		10M CK(N) to XDATSNKN				-	-	-		
		10M CK(N+1) to XNABLSRCN+1				-	-	-		
		10M CK(N+1) to XDATSRCN+1				-	-	-		
		10M CK(N+1) to XNABLSNKN+1				-	-	-		
12	th	10M CK(N+1) to XDATSNKN+1	10	10	10	-	-	-	nS	12

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6.4 Module function. The MIL-STD-1397 type E serial interface adapter module performs Manchester II encoding, decoding, and biasing. The module contains pulse transformers, delay lines, control drivers, control receiver and Manchester II encoder/decoder, receiver/transmitter circuits for channel N and N+1. Additionally, the module contains a receiver flip-flop and a timeout clock.

6.4.1 Serial North Atlantic Treaty Organization (NATO) Encoder/Decoder, Receiver/Transmitter (SNERT). The SNERT integrated circuit performs serial interface Manchester II encoding/decoding and transceiver functions. This circuit provides the two independent transceiver functions required (output and input), for one full-duplex, digital, serial Input/Output (I/O) channel. The encoder/transmitter portion of the SNERT chip accepts Transistor-Transistor Logic (TTL) clock timing, Non-Return-to-Zero (NRZ) data, and driver enable signals from the channel control logic, then generates the Manchester code signal for serial bit transmission at 10 megabits per second. The receiver/decoder portion of the chip receives the Manchester code signal from the I/O cable (via a signal transformer), decodes the serial data and clock information, then outputs them as TTL compatible signals. This circuit also has the capability to loopback data and control signals from side A to side B and vice versa. This provides I/O channel test capability at the system level without disconnecting the I/O cables and reconnecting a jumper cable for test.

6.4.2 Delay line. The 10-75 nS delay line decodes the incoming Manchester encoded data. The delay line is an analog delay terminated by 100 ohms at 0.45 V. The delay line driver signal (RCV OUTPUT) is an analog signal. The SNERT is designed to accept a 0.8 V amplitude signal with the dc level shifting with both temperature and supply voltages. These delay lines are physically situated close to the SNERT to minimize the channel crosstalk.

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6.4.3 Description of I/O pins.

Signal name	Pin number	I/O	Description
20M CK(N+1)	8	I	20 MHZ CLK (CHAN N+1)
XNABLSRCN	11	I	TRANS ENBL SOURCE N
XNABLSNKN	12	I	TRANS ENBL SINK N
XDATSNKN	18	I	TRANS DATA SINK N
XNABLSRCN+1	28	I	TRANS ENBL SOURCE N+1
XDATSNKN+1	29	I	TRANS DATA SINK N+1
XNABLSNKN+1	62	I	TRANS ENBL SINK N+1
CLR CNTR	66	I	CLR CNTR
20M CK(N)	67	I	20 MHZ CLK (CHAN N)
LPBKMDEN+1	74	I	LOOPBACK MODE N+1
XDATSRCN+1	78	I	TRANS DATA SOURCE N+1
LPBKMDEN	80	I	LOOPBACK MODE N
XDATSRCN	99	I	TRANS DATA SOURCE N
PWR MACLR	58	I	PWR MA CLR
RCVCKSNKN	9	O	RCV CLK SINK N
10M CK(N)	13	O	10 MHZ CLK (CHAN N)
RCVDTSNKN	15	O	RCV DATA SINK N
RCVENSNKN	17	O	RCV ENV SINK N
RCVDTSNKN+1	21	O	RCV DATA SINK N+1
RCVCKSNKN+1	26	O	RCV CLK SINK N+1

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RCVENSNKN+1	30	O	RCV ENV SINK N+1
RCVCKSRCN	52	O	RCV CLK SOURCE N
RCVDTSRCN	61	O	RCV DATA SOURCE N
RCVENSRCN	63	O	RCV ENV SOURCE N
RCVDTSRCN+1	76	O	RCV DATA SOURCE N+1
RCVENSRCN+1	77	O	RCV ENV SOURCE N+1
TMOUTCK N	79	O	TIMEOUT CLK (CHAN N)
10M CK(N+1)	87	O	10 MHZ CLK (CHAN N+1)
TMOUTCK N+1	95	O	TIMEOUT CLK (CHAN N+1)
RCVCKSRCN+1	96	O	RCV CLK SOURCE N+1
RTN SNK N	22	O	RTN SINK CHAN N
SIG SNK N	23	O	SIGNAL SINK CHAN N
RTN SRC N	35	O	RTN SOURCE CHAN N
SIG SRC N	36	O	SIGNAL SOURCE CHAN N
RTN SRC N+1	70	O	RTN SOURCE CHAN N+1
SIG SRC N+1	71	O	SIGNAL SOURCE CHAN N+1
RTN SNK N+1	82	O	RTN SINK CHAN N+1
SIG SNK N+1	83	O	SIGNAL SINK CHAN N+1
TYPE E N	37	I	TYPE E
TYPE E N+1	41	I	TYPE E N

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Surface Warfare Center, Crane Division (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0043)

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5C Initial limits Min	25 ± 5C Initial limits Max	Units	Notes
A1	Isolation, frame	C,2,3, 4,5,6, 7,14, 16,19, 24,27, 32,33, 34,38, 39,42, 43,44, 45,46, 47,48, 49,50, 53,54, 55,56, 57,59, 64,65, 68,69, 72,73, 84,85, 86,88, 89,91, 92,93, 94,97, 98,100	Frame	10	INF	MOhm	1-9

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5C Initial limits Min	25 ± 5C Initial limits Max	Units	Notes
A2	Isolation, used to unused pins	C	2,3,4, 5,6,7, 14,16, 19,24, 27,32, 33,34, 38,39, 42,43, 44,45, 46,47, 48,49, 50,53, 54,55, 56,57, 59,64, 65,68, 69,72, 73,84, 85,86, 88,89, 91,92, 93,94, 97,98, 100	10	INF	MOhm	1-9
A3-A50	Isolation, unused to unused pins	See note 4		10	INF	MOhm	1-9

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A50
Isolation measurement	NA	± 500 KOhm	A1-A50

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 2,3,4,5,6,7,14,16,19, 24,27,32,33,34,38,39,42,43,44,45,46,47,48,49,50,53,54,55,56,57, 59,64,65,68,69,72,73,84,85,86,88,89,91,92,93,94,97,98 and 100. When testing for unused to unused pin isolation, connect one unused pin to A and the remaining unused pins to B.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 μA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

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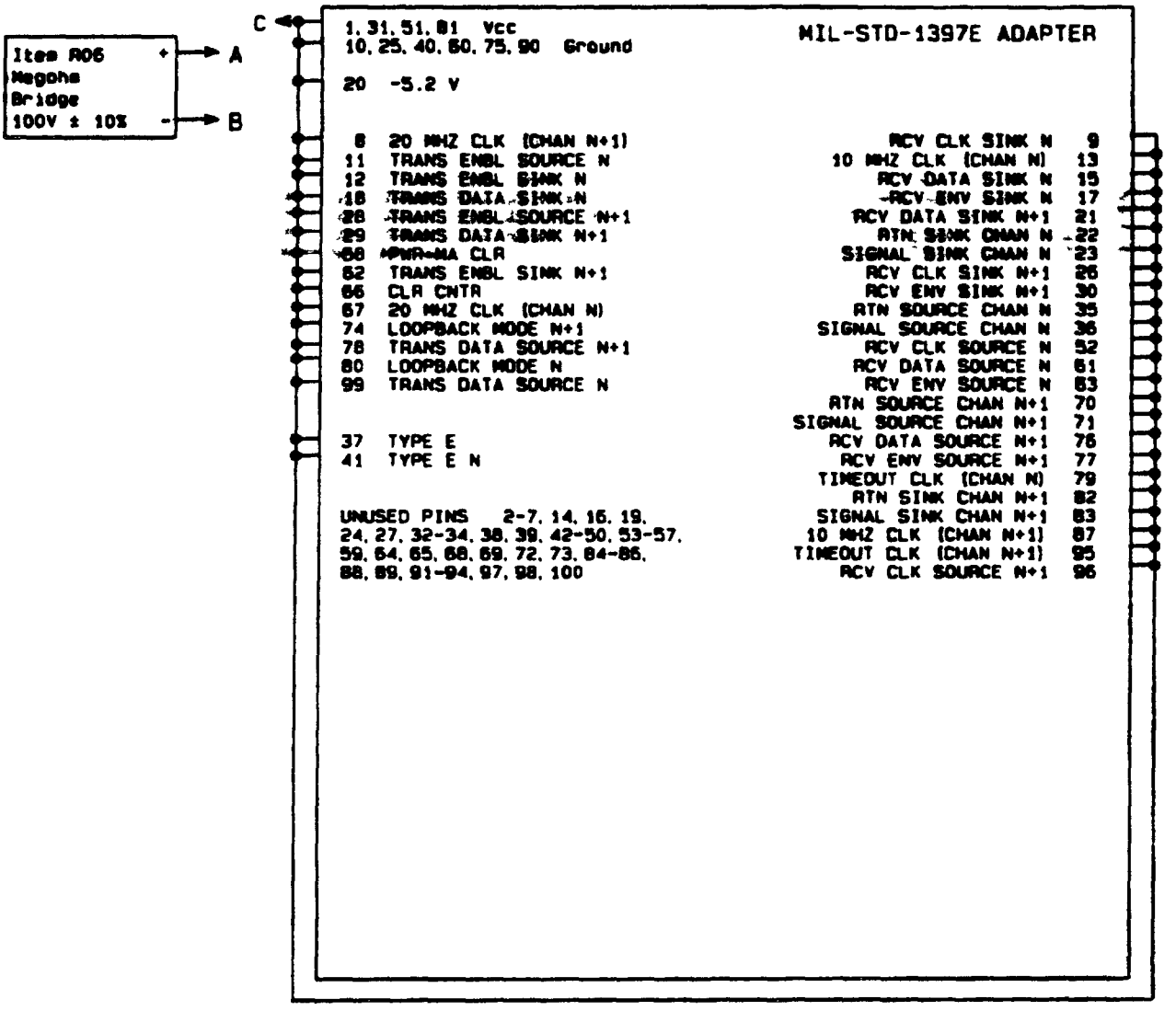


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81				
B7	Continuity, TYPE E N to TYPE E N+1	37	41				
B8	Continuity TYPE E N to GND	37	10	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B8

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

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NOTES: - Continued.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1, 31, 51, 81 Vcc
10, 25, 40, 60, 75, 90 Ground

20 -5.2 V

MIL-STD-1397E ADAPTER

8	20 MHZ CLK (CHAN N+1)	RCV CLK SINK N	9
11	TRANS ENBL SOURCE N	10 MHZ CLK (CHAN N)	13
12	TRANS ENBL SINK N	RCV DATA SINK N	15
18	TRANS DATA SINK N	RCV ENV SINK N	17
28	TRANS ENBL SOURCE N+1	RCV DATA SINK N+1	21
29	TRANS DATA SINK N+1	RTN SINK CHAN N	22
58	PWR MA CLR	SIGNAL SINK CHAN N	23
62	TRANS ENBL SINK N+1	RCV CLK SINK N+1	26
66	CLR CNTR	RCV ENV SINK N+1	30
67	20 MHZ CLK (CHAN N)	RTN SOURCE CHAN N	35
74	LOOPBACK MODE N+1	SIGNAL SOURCE CHAN N	36
78	TRANS DATA SOURCE N+1	RCV CLK SOURCE N	52
80	LOOPBACK MODE N	RCV DATA SOURCE N	61
99	TRANS DATA SOURCE N	RCV ENV SOURCE N	63
		RTN SOURCE CHAN N+1	70
		SIGNAL SOURCE CHAN N+1	71
37	TYPE E	RCV DATA SOURCE N+1	76
41	TYPE E N	RCV ENV SOURCE N+1	77
		TIMEOUT CLK (CHAN N)	79
		RTN SINK CHAN N+1	82
UNUSED PINS	2-7, 14, 16, 19,	SIGNAL SINK CHAN N+1	83
	24, 27, 32-34, 38, 39, 42-50, 53-57,	10 MHZ CLK (CHAN N+1)	87
	59, 64, 65, 68, 69, 72, 73, 84-86,	TIMEOUT CLK (CHAN N+1)	95
	88, 89, 91-94, 97, 98, 100	RCV CLK SOURCE N+1	96

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6
C	V2	7.0	10,25, 40,60, 75,90	20	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
(V2)	-7.0 V	+70 mV,-0 mV	C
Power supply current limit, (Vcc)	2306 mA	NA	C
(V2)	-466.9mA	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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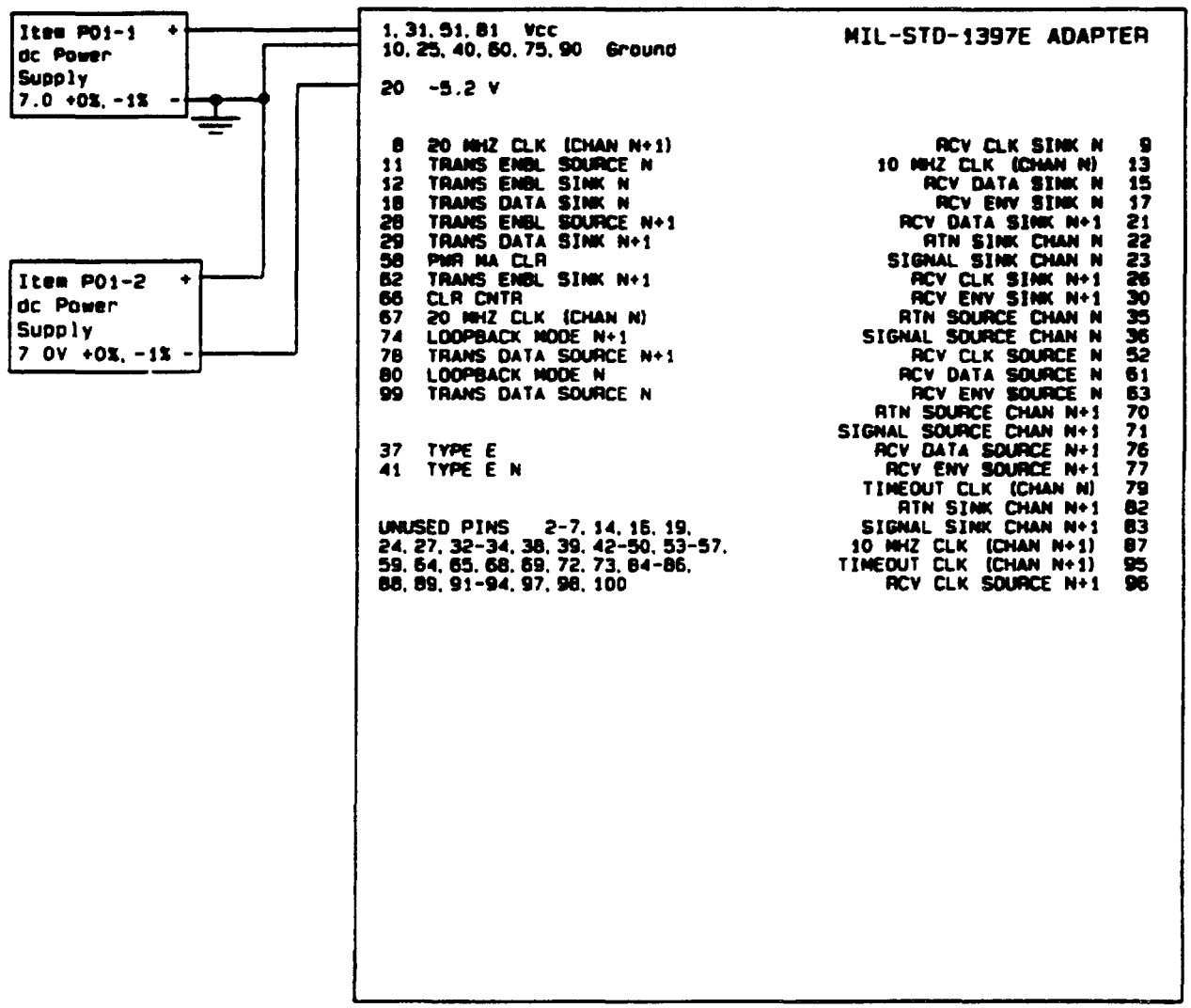


FIGURE 8. Power supply stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	20M CK(N+1)	8	5.5	-0.02	1.02	mA	1-3
D2	XNABLSRCN	11		0.0	0.1		
D3	XNABLSNKN	12					
D4	XDATSNKN	18					
D5	XNABLSRCN+1	28					
D6	XDATSNKN+1	29		0.0	0.1		
D7	PWR MACLR	58		-0.58	1.14		
D8	XNABLSNKN+1	62		0.0	0.1		
D9	CLR CNTR	66		-0.02	0.22		
D10	20M CK(N)	67		-0.02	1.02		
D11	LPBKMDEN+1	74		-0.02	1.02		
D12	XDATSRCN+1	78		0.0	0.1		
D13	LPBKMDEN	80		-0.02	1.02		
D14	XDATSRCN	99	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	5.5 V -5.7 V	± 55 mV ± 57 mV	D1-D14 D1-D14
Power supply current limit, (Vcc) (V2)	1845 mA -373.5 mA	NA NA	D1-D14 D1-D14
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D14
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	D1-D14 D1-D14
Input current measurement	NA NA	± 52 µA ± 5 µA ± 86 µA ± 5 µA ± 12 µA ± 52 µA ± 5 µA ± 52 µA ± 5 µA	D1 D2-D6 D7 D8 D9 D10,D11 D12 D13 D14

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5C Initial limits Min	25 ± 5C Initial limits Max	Units	Notes
E1	RCVCKSNKN	9	0.0	-100.0	-40.0	mA	1-4
E2	10M CK(N)	13					
E3	RCVDTSNKN	15					
E4	RCVENSNKN	17					
E5	RCVDTSNKN+1	21					
E6	RCVCKSNKN+1	26					
E7	RCVENSNKN+1	30					
E8	RCVCKSRCN	52					
E9	RCVDTSRCN	61					
E10	RCVENSRCN	63					
E11	RCVDTSRCN+1	76					
E12	RCVENSRCN+1	77		-100.0	-40.0		
E13	TMOUTCK N	79		-85.0	-15.0		
E14	10M CK(N+1)	87		-100.0	-40.0		
E15	TMOUTCK N+1	95		-85.0	-15.0		
E16	RCVCKSRCN+1	96	0.0	-100.0	-40.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E16
(V2)	-5.7 V	± 57 mV	E1-E16
Power supply current limit, (Vcc)	1845 mA	NA	E1-E16
(V2)	-373.5 mA	NA	E1-E16
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E16
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E16
low, (VL)	0.0 V	± 100 mV	E1-E16
Output current measurement	NA	± 3.0 mA	E1-E12
		± 3.5 mA	E13
		± 3.0 mA	E14
		± 3.5 mA	E15
	NA	± 3.0 mA	E16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	20M CK(N+1)	8	2.4	-3.04	-0.5	mA	1-3
F2	XNABLSRCN	11		-0.4	0.02		
F3	XNABLSNKN	12					
F4	XDATSNKN	18					
F5	XNABLSRCN+1	28					
F6	XDATSNKN+1	29		-0.4	0.02		
F7	PWR MACLR	58		-6.64	-0.35		
F8	XNABLSNKN+1	62		-0.4	0.02		
F9	CLR CNTR	66		-1.64	-0.51		
F10	20M CK(N)	67		-3.04	-0.5		
F11	LPBKMDEN+1	74		-3.04	-0.5		
F12	XDATSRCN+1	78		-0.4	0.02		
F13	LPBKMDEN	80		-3.04	-0.5		
F14	XDATSRCN	99	2.4	-0.4	0.02	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	5.5 V -5.7 V	± 55 mV ± 57 mV	F1-F14 F1-F14
Power supply current limit, (Vcc) (V2)	1845 mA -373.5 mA	NA NA	F1-F14 F1-F14
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F14
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	F1-F14 F1-F14
Input current measurement	NA NA	± 127.0 µA ± 21.0 µA ± 314.5 µA ± 21.0 µA ± 56.5 µA ± 127.0 µA ± 21.0 µA ± 127.0 µA ± 6.0 µA	F1 F2-F6 F7 F8 F9 F10,F11 F12 F13 F14

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	20M CK(N+1)	8	0.4	-3.04	-0.5	mA	1-3
G2	XNABLSRCN	11		-0.4	0.02		
G3	XNABLSNKN	12					
G4	XDATSNKN	18					
G5	XNABLSRCN+1	28					
G6	XDATSNKN+1	29		-0.4	0.02		
G7	PWR MACLR	58		-6.64	-0.35		
G8	XNABLSNKN+1	62		-0.4	0.02		
G9	CLR CNTR	66		-1.64	-0.51		
G10	20M CK(N)	67		-3.04	-0.5		
G11	LPBKMDEN+1	74		-3.04	-0.5		
G12	XDATSRCN+1	78		-0.4	0.02		
G13	LPBKMDEN	80		-3.04	-0.5		
G14	XDATSRCN	99	0.4	-0.4	0.02	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	5.5 V -5.7 V	± 55 mV ± 57 mV	G1-G14 G1-G14
Power supply current limit, (Vcc) (V2)	1845 mA -373.5 mA	NA NA	G1-G14 G1-G14
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G14
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G14 G1-G14
Input current measurement	NA NA	± 127.0 µA ± 21.0 µA ± 314.5 µA ± 21.0 µA ± 56.5 µA ± 127.0 µA ± 21.0 µA ± 127.0 µA ± 21.0 µA	G1 G2-G6 G7 G8 G9 G10,G11 G12 G13 G14

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5C Initial limits Min	25 ± 5C Initial limits Max	Units	Notes
H1	RCVCKSNKN	9	-0.4	2.4	4.5	V	1-4
H2	10M CK(N)	13					
H3	RCVDTSNKN	15					
H4	RCVENSNKN	17					
H5	RCVDTSNKN+1	21					
H6	RCVCKSNKN+1	26					
H7	RCVENSNKN+1	30					
H8	RCVCKSRCN	52					
H9	RCVDTSRCN	61					
H10	RCVENSRCN	63					
H11	RCVDTSRCN+1	76					
H12	RCVENSRCN+1	77	-0.4				
H13	TMOUTCK N	79	-0.16				
H14	10M CK(N+1)	87	-0.4				
H15	TMOUTCK N+1	95	-0.16				
H16	RCVCKSRCN+1	96	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	4.5 V -4.7 V	± 45 mV ± 47 mV	H1-H16 H1-H16
Power supply current limit, (Vcc) (V2)	1845 mA -373.5 mA	NA NA	H1-H16 H1-H16
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA -0.16 mA -0.4 mA	± 40 µA ± 16 µA ± 40 µA ± 16 µA ± 40 µA	H1-H12 H13 H14 H15 H16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H16 H1-H16
Output voltage measurement	NA	± 105 mV	H1-H16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

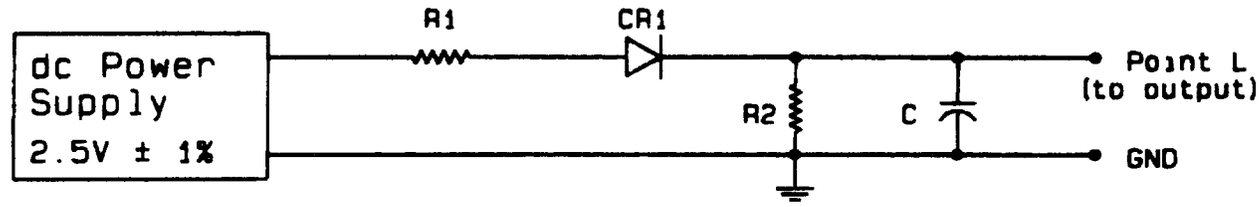
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

    (a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

    (b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

    (c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

    (d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
9,13,15,17,21, 26,30,52,61,63, 76,77,87,96	6200	6.0	0.5
79,95	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5C Initial limits Min	Max	Units	Notes
J1	RCVCKSNKN	9	6.0	0.0	0.5	V	1-4
J2	10M CK(N)	13					
J3	RCVDTSNKN	15					
J4	RCVENSNKN	17					
J5	RCVDTSNKN+1	21					
J6	RCVCKSNKN+1	26					
J7	RCVENSNKN+1	30					
J8	RCVCKSRCN	52					
J9	RCVDTSRCN	61					
J10	RCVENSRCN	63					
J11	RCVDTSRCN+1	76					
J12	RCVENSRCN+1	77	6.0				
J13	TMOUTCK N	79	1.6				
J14	10M CK(N+1)	87	6.0				
J15	TMOUTCK N+1	95	1.6				
J16	RCVCKSRCN+1	96	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J16
(V2)	-4.7 V	± 47 mV	J1-J16
Power supply current limit, (Vcc)	1845 mA	NA	J1-J16
(V2)	-373.5 mA	NA	J1-J16
Forcing current, (Iol)	6.0 mA	± 600 µA	J1-J12
	1.6 mA	± 160 µA	J13
	6.0 mA	± 600 µA	J14
	1.6 mA	± 160 µA	J15
	6.0 mA	± 600 µA	J16
Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J16
low, (VL)	0.0 V	± 100 mV	J1-J16
Output voltage measurement	NA	± 25 mV	J1-J16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5C Initial limits Min	Max	Units	Notes
L1	XNABLSRCN	11	-18.0	-1.5	0.0	V	1-3
L2	XNABLSNKN	12					
L3	XDATSNKN	18					
L4	XNABLSRCN+1	28					
L5	XDATSNKN+1	29					
L6	XNABLSNKN+1	62					
L7	XDATSRCN+1	78					
L8	XDATSRCN	99	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L8
(V2)	Off	NA	L1-L8
Power supply current limit, (Vcc)	NA	NA	L1-L8
(V2)	NA	NA	L1-L8
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L8

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L8 L1-L8
Input voltage measurement	NA	± 75 mV	L1-L8

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
M1	PSFC	1	2.3	4.3	μF	1-4
M2	PSFC	51	4.5	8.3	μF	1-4
M3	PSFC	20	2.4	4.6	μF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1-M3
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1-M3
Capacitance measurement	NA | NA	± 100 nF ± 190 nF ± 110 nF	M1 M2 M3

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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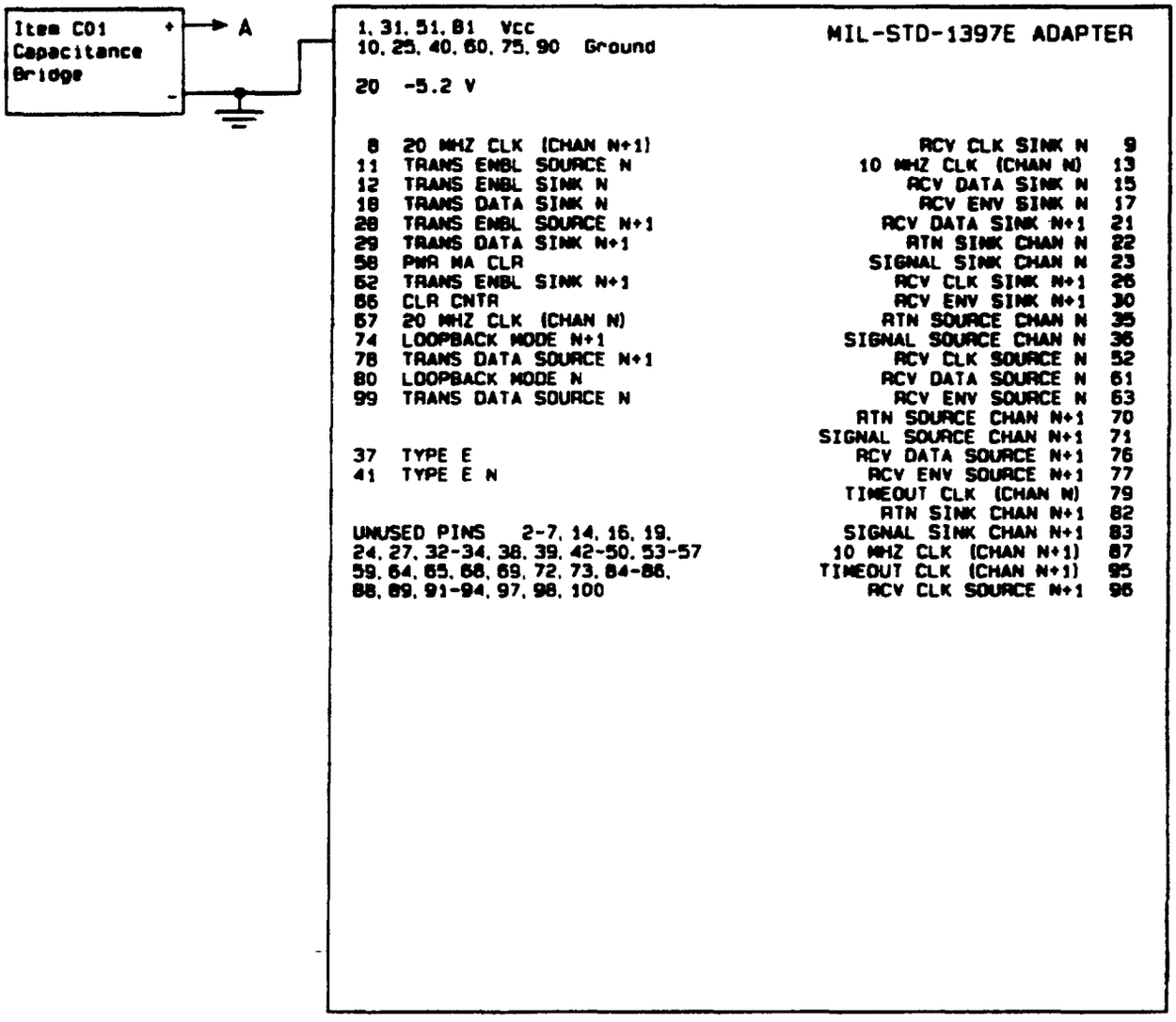


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5C Initial limits Min	Max	Units	Notes
N1	Icc	1,31, 51,81	520	1230	mA	1-3
N2	Icc	20	-249	-100	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1,N2
(V2)	-5.7 V	± 57 mV	N1,N2
Power supply current limit, (Vcc)	1845 mA	NA	N1,N2
(V2)	-373.5 mA	NA	N1,N2
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1,N2
low, (VL)	0.0 V	± 100 mV	N1,N2
Power supply current measurement	NA	± 35.5 mA	N1
	NA	± 7.5 mA	N2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (100 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (100 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2) (Vcc) (V2)	4.5 V -4.7 V 5.5 V -5.7 V	± 45 mV ± 47 mV ± 55 mV ± 57 mV	P1 P1 P2 P2
Power supply current limit, (Vcc) (V2)	1845mA -373.5mA	NA NA	P1,P2 P1,P2
Pattern voltage high, (VH) low, (VL)	2.4 V 0.4 V	± 100 mV ± 100 mV	P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	NA | NA	± 105 mV ± 155 mV ± 25 mV	P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

For pins 8,9,11,12,13,15,17,18,21,26,28,29,30,37,41,52,58,61,62, 63,66,67,74,76,77,78,79,80,87,95,96,99:

High level: 2.4 V ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ 0.5 V

For pins 22,23,35,36,70,71,82,83:

High level: 0.0V ≤ Voh ≤ 0.3
Low level: -0.3V ≤ Vol ≤ 0.0V

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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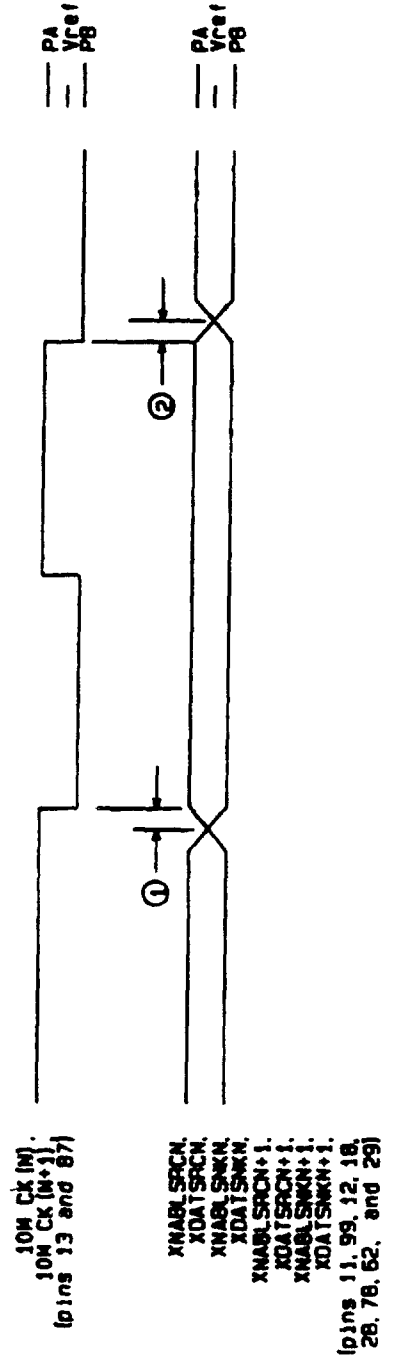


FIGURE 11. Input timing parameters for table XIV.

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NOTES:

1. (1) XNABLSRCN to 10M CK(N). Minimum setup time: 12 nS.
2. (1) XDATSRCN to 10M CK(N). Minimum setup time: 12 nS.
3. (1) XNABLSNKN to 10M CK(N). Minimum setup time: 12 nS.
4. (1) XDATSNKN to 10M CK(N). Minimum setup time: 12 nS.
5. (1) XNABLSRCN+1 to 10M CK(N+1). Minimum setup time: 12 nS.
6. (1) XDATSRCN+1 to 10M CK(N+1). Minimum setup time: 12 nS.
7. (1) XNABLSNKN+1 to 10M CK(N+1). Minimum setup time: 12 nS.
8. (1) XDATSNKN+1 to 10M CK(N+1). Minimum setup time: 12 nS.
9. (2) 10M CK(N) to XNABLSRCN. Minimum setup time: 10 nS.
10. (2) 10M CK(N) to XDATSRCN. Minimum setup time: 10 nS.
11. (2) 10M CK(N) to XNABLSNKN. Minimum setup time: 10 nS.
12. (2) 10M CK(N) to XDATSNKN. Minimum setup time: 10 nS.
13. (2) 10M CK(N+1) to XNABLSRCN+1. Minimum setup time: 10 nS.
14. (2) 10M CK(N+1) to XDATSRCN+1. Minimum setup time: 10 nS.
15. (2) 10M CK(N+1) to XNABLSNKN+1. Minimum setup time: 10 nS.
16. (2) 10M CK(N+1) to XDATSNKN+1. Minimum setup time: 10 nS.

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TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (100 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (100 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (100 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (100 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Q1,Q3
(V2)	-4.7 V	± 47 mV	Q1-Q3
(Vcc)	5.5 V	± 55 mV	Q2,Q4
(V2)	-5.7 V	± 57 mV	Q2,Q4
Power supply current limit, (Vcc)	1845mA	NA	Q1-Q4
(V2)	-373.5mA	NA	Q1-Q4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The pattern voltages shall be as follows:

Pin number	Pattern voltage				
	High (VH)	Accuracy	Low (VL)	Accuracy	Test condition
8,11,12,18, 28,29,58,62, 66,67,74,78, 80,99	2.4V	± 100 mV	0.4V	± 100 mV	Q1-Q4

(c) The output compare levels shall be as follows.

Pin number	Output compare levels				
	High level	Accuracy	Low level	Accurcy	Test condition
9,13,15,17, 21,26,30,37, 41,52,61,63, 76,77,79,87, 95,96	Vref ≤ Voh ≤ Vcc	± 100 mV	0.0V ≤ Vol ≤ Vref	± 100 mV	Q1-Q4
22,23,35,36, 70,71,82,83	Vref ≤ Voh ≤ 0.3V	± 100 mV	-0.3V ≤ Vol ≤ Vref	± 100 mV	Q1-Q4

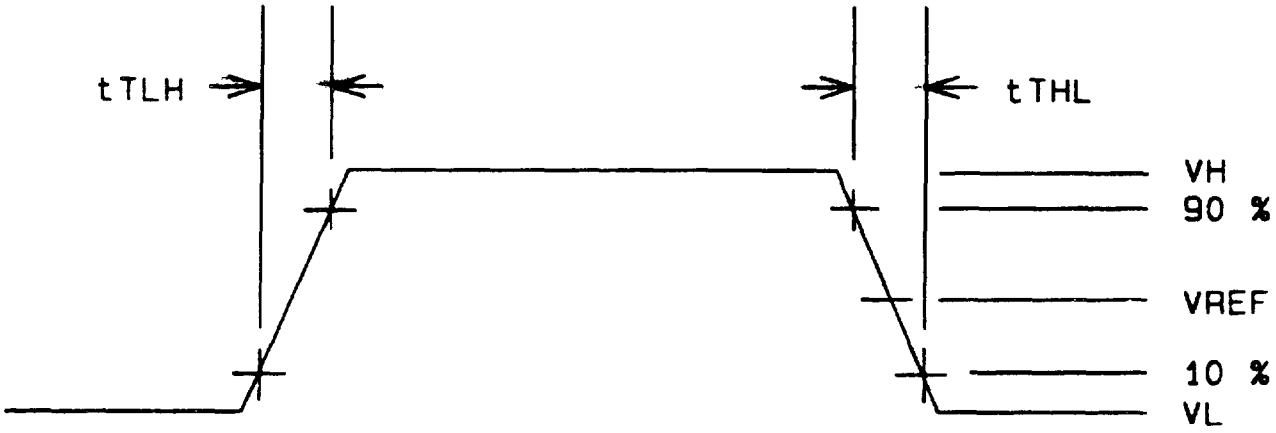
(d) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

4. I/O waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:

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NOTES: - Continued.



Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL ≤ 10 nS.

(c) Vref = 1.6 V for pins: 8,9,11,12,13,15,17,18,21,26,28,29, 30,37,41,52,58,61,62,63,66,67,74,76,77,78,79,80,87,95,96,99.

(d) Vref = 0.0V for pins: 22,23,35,36,70,71,82,83.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	RTN SRC N to RCVENSRCN	-	-	-	56	56	56	nS	12
		SIG SRC N to RCVENSRCN	-	-	-					
		RTN SRC N+1 to RCVENSRCN+1	-	-	-					
1		SIG SRC N+1 to RCVENSRCN+1	-	-	-	56	56	56		
2		RTN SRC N to RCVENSRCN	-	-	-	206	206	206		
		SIG SRC N to RCVENSRCN	-	-	-					
		RTN SRC N+1 to RCVENSRCN+1	-	-	-					
2		SIG SRC N+1 to RCVENSRCN+1	-	-	-	206	206	206		
3		RTN SRC N to RCVCKSRCN	-	-	-	176	176	176		
3	tpd	SIG SRC N to RCVCKSRCN	-	-	-	176	176	176	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	RTN SRC N+1 to RCVCKSRCN+1	-	-	-	176	176	176	nS	12
3		SIG SRC N+1 to RCVCKSRCN+1	-	-	-	176	176	176		
4		RTN SNK N to RCVENSNKN	-	-	-	56	56	56		
		SIG SNK N to RCVENSNKN	-	-	-					
		RTN SNK N+1 to RCVENSNKN+1	-	-	-					
4		SIG SNK N+1 to RCVENSNKN+1	-	-	-	56	56	56		
5		RTN SNK N to RCVENSNKN	-	-	-	206	206	206		
		SIG SNK N to RCVENSNKN	-	-	-					
		RTN SNK N+1 to RCVENSNKN+1	-	-	-					
5	tpd	SIG SNK N+1 to RCVENSNKN+1	-	-	-	206	206	206	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	RTN SNK N to RCVCKSNKN	-	-	-	176	176	176	nS	12
		SIG SNK N to RCVCKSNKN	-	-	-					
		RTN SNK N+1 to RCVCKSNKN+1	-	-	-					
6		SIG SNK N+1 to RCVCKSNKN+1	-	-	-	176	176	176		
7		20M CK(N) to 10M CK(N)	-	-	-	50	50	50		
7		20M CK(N+1) to 10M CK(N+1)	-	-	-	50	50	50		
8		20M CK(N) to 10M CK(N)	-	-	-	59	59	59		
8		20M CK(N+1) to 10M CK(N+1)	-	-	-	59	59	59		
9		20M CK(N) to TMOUTCK N	-	-	-	84	84	84		
9		20M CK(N+1) to TMOUTCK N+1	-	-	-					
10	tpd	20M CK(N) to TMOUTCK N	-	-	-	84	84	84	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tpd	20M CK(N+1) to TMOUTCK N+1	-	-	-	84	84	84	nS	12
11	tsu	XNABLSRCN to 10M CK(N)	12	12	12	-	-	-		
		XDATSRCN to 10M CK(N)				-	-	-		
		XNABLSNKN to 10M CK(N)				-	-	-		
		XDATSNKN to 10M CK(N)				-	-	-		
		XNABLSRCN+1 to 10M CK(N+1)				-	-	-		
		XDATSRCN+1 to 10M CK(N+1)				-	-	-		
		XNABLSNKN+1 to 10M CK(N+1)				-	-	-		
11	tsu	XDATSNKN+1 to 10M CK(N+1)	12	12	12	-	-	-		
12	th	10M CK(N) to XNABLSRCN	10	10	10	-	-	-		
12	th	10M CK(N) to XDATSRCN	10	10	10	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	th	10M CK(N) to XNABLSNKN	10	10	10	-	-	-	nS	12
		10M CK(N) to XDATSNKN				-	-	-		
		10M CK(N+1) to XNABLSRCN+1				-	-	-		
		10M CK(N+1) to XDATSRCN+1				-	-	-		
		10M CK(N+1) to XNABLSNKN+1				-	-	-		
12	th	10M CK(N+1) to XDATSNKN+1	10	10	10	-	-	-	nS	12

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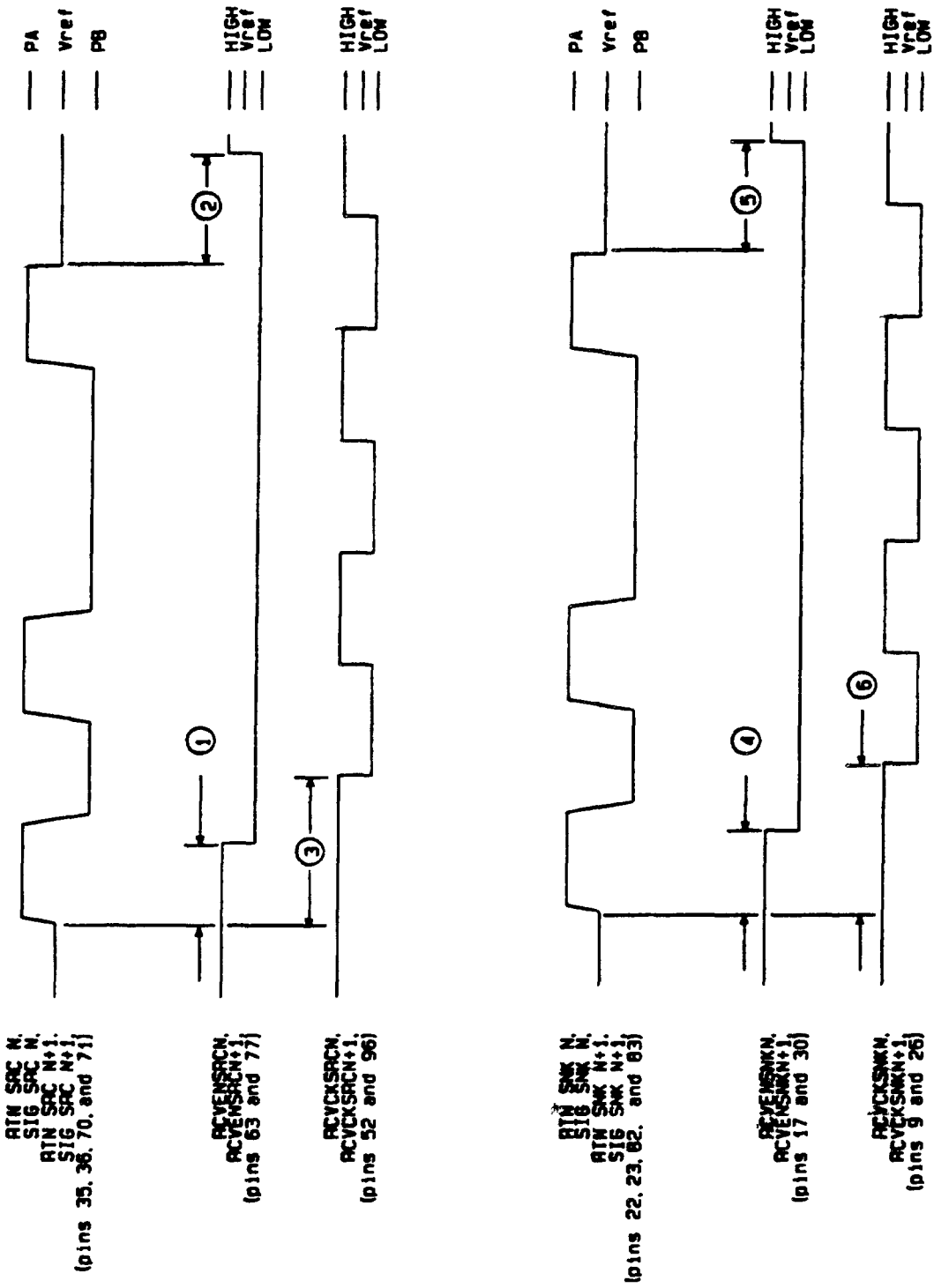


FIGURE 12a. Dynamic waveforms for table XVI.

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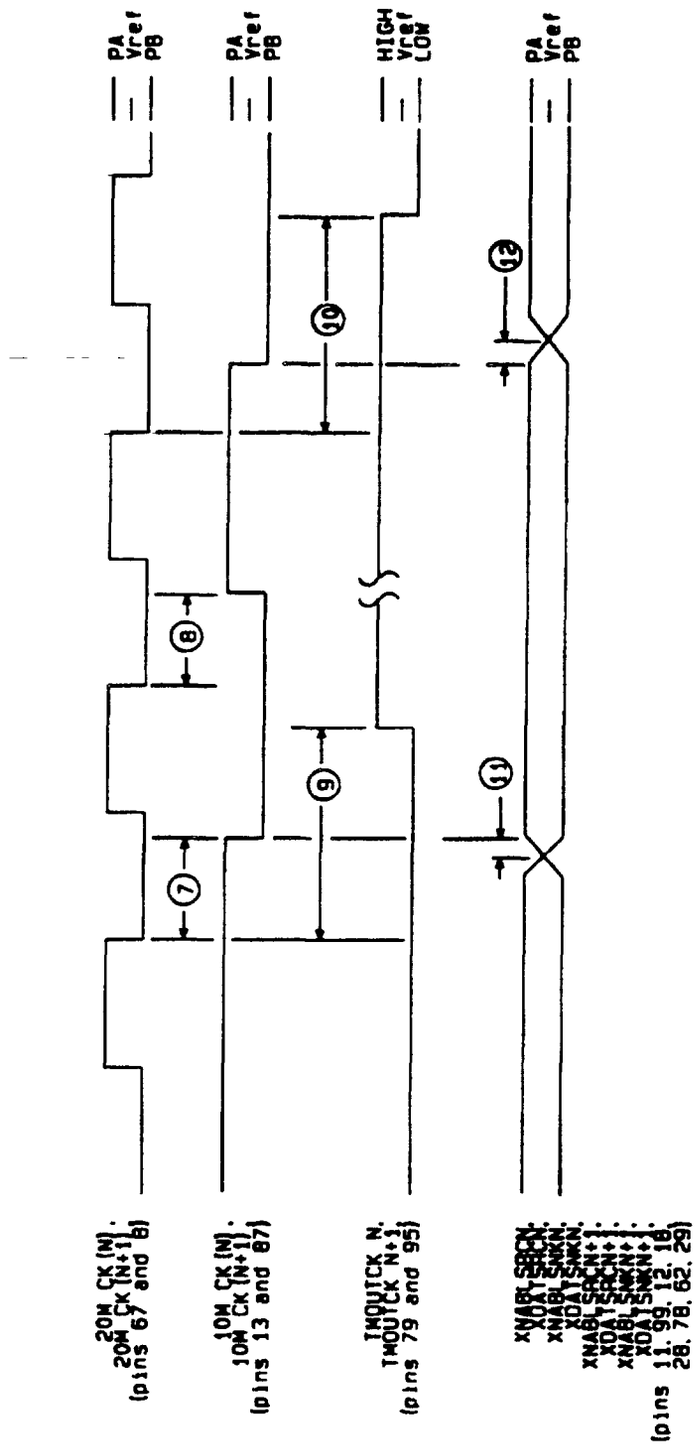


FIGURE 12b. Dynamic waveforms for table XVI.

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TABLE XVII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5C Initial limits Min	25 ± 5C Initial limits Max	Units	Notes
R1	20M CK(N+1)	8	2	75	pF	1-4
R2	XNABLSRCN	11				
R3	XNABLSNKN	12				
R4	XDATSNKN	18				
R5	XNABLSRCN+1	28				
R6	XDATSNKN+1	29	2	75		
R7	PWR MACLR	58	10	235		
R8	XNABLSNKN+1	62	2	100		
R9	CLR CNTR	66	2	125		
R10	20M CK(N)	67	4	75		
R11	LPBKMDEN+1	74	4	100		
R12	XDATSRCN+1	78	2	75		
R13	LPBKMDEN	80	4	75		
R14	XDATSRCN	99	2	125	pF	1-4

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NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R14
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R14
Capacitance measurement	NA NA	± 3.65 pF ± 11.2 pF ± 4.9 pF ± 6.15 pF ± 3.55 pF ± 4.8 pF ± 3.65 pF ± 3.55 pF ± 6.15 pF	R1-R6 R7 R8 R9 R10 R11 R12 R13 R14

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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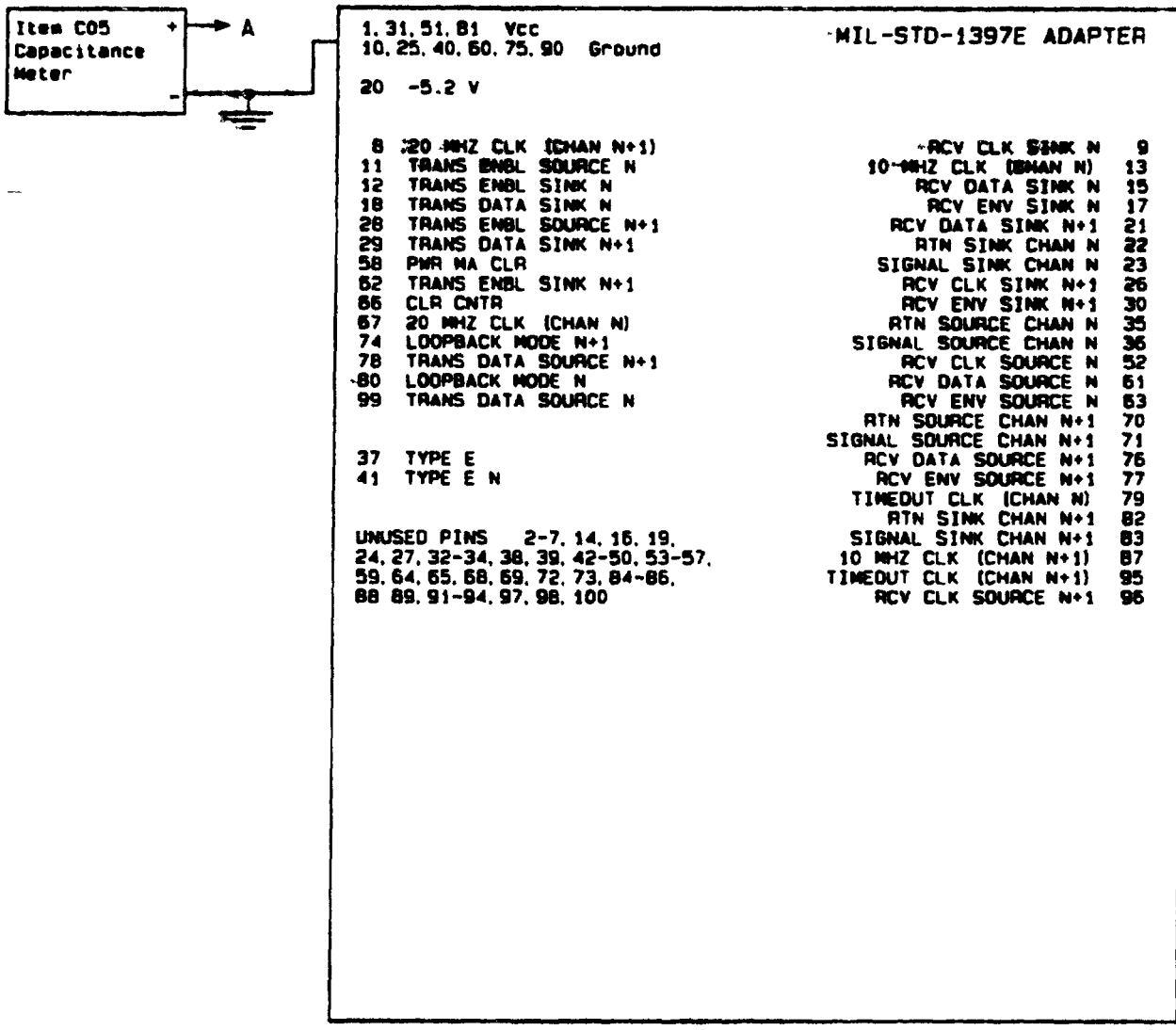


FIGURE 13. Input capacitance test circuit for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	20	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	5.0 V -5.2 V	± 500 mV ± 52 mV	T1,T2 T1,T2
Power supply current limit, (Vcc) (V2)	1845 mA -373.5 mA	NA NA	T1,T2 T1,T2
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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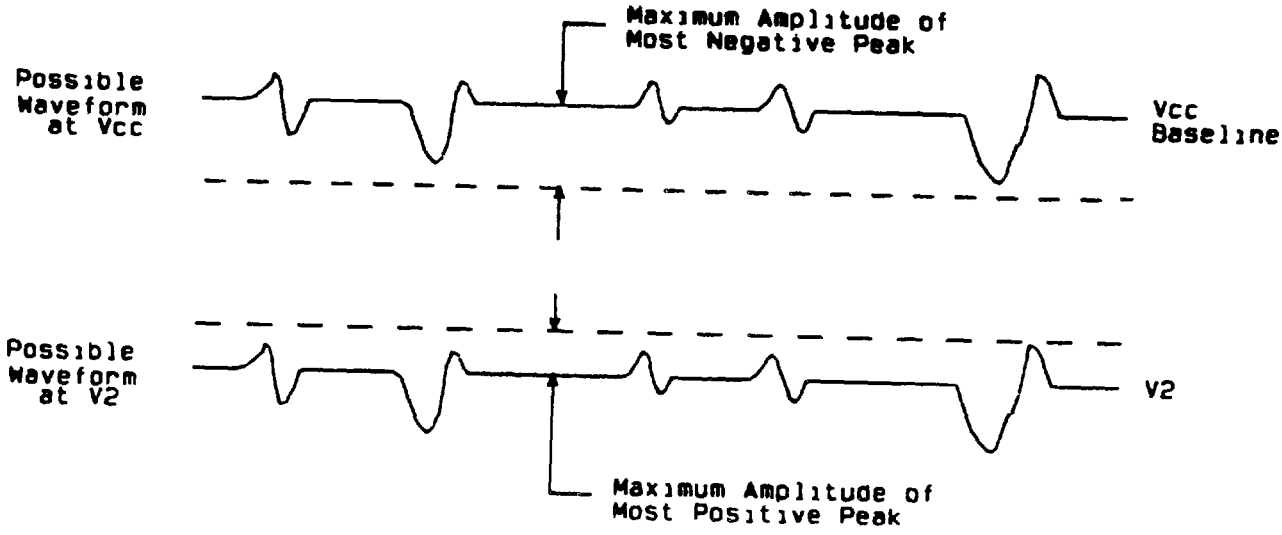
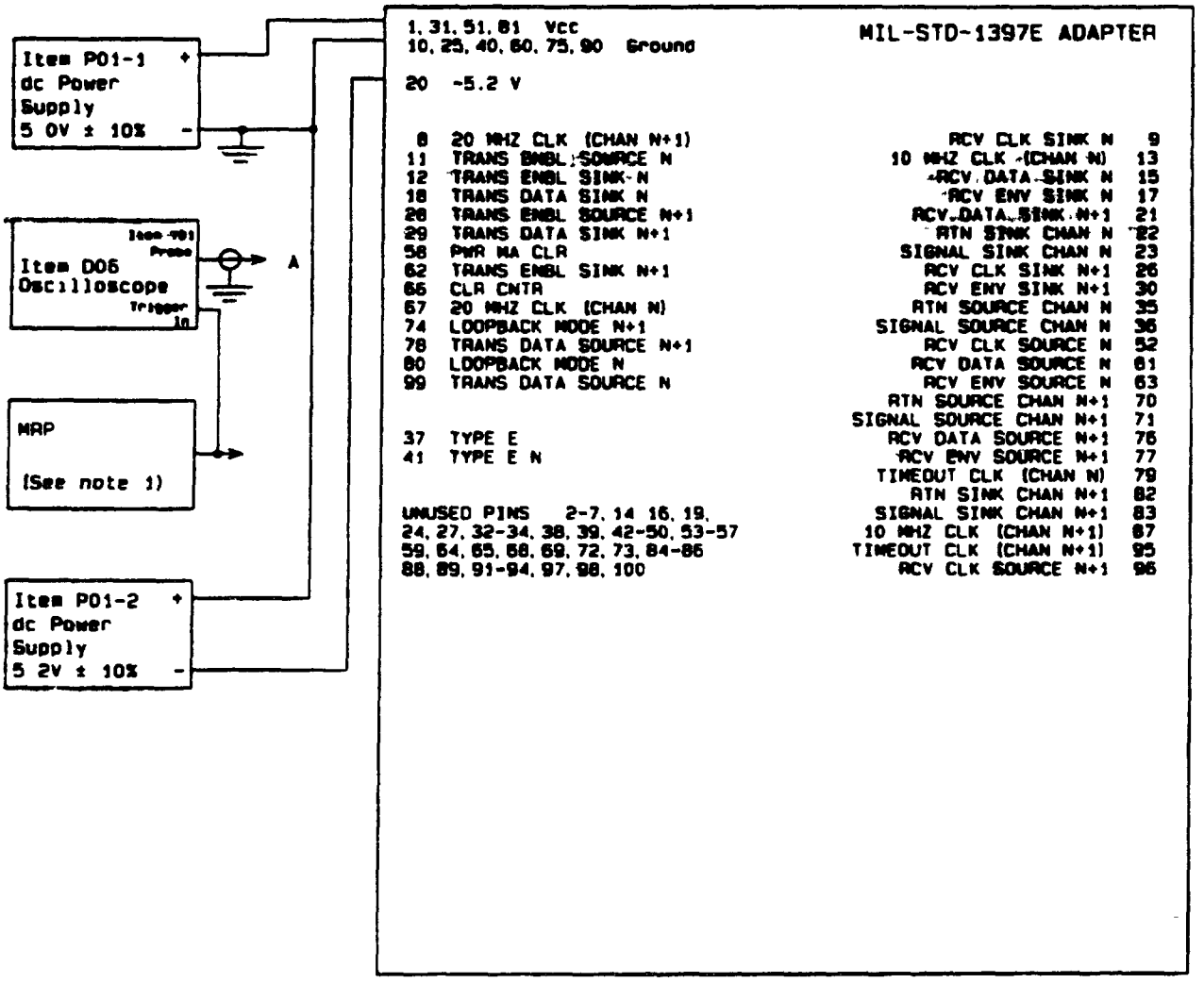


FIGURE 14. Power supply transient amplitude waveform for table XVIII.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XVIII.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-M-28787/414	2 DOCUMENT DATE (YYMMDD) 1994 March 23

3. DOCUMENT TITLE Modules, Standard Electronic, MIL-STD-1397 Type E Serial Interface Adapter Module, Key Code ATC

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME Technical Point of Contact (TPOC): Mr. Joseph P. Toy (Code 603A)	b TELEPHONE (Include Area Code) (1) Commercial TPOC: 812-854-5993	(2) AUTOVON 88-482-5993
c ADDRESS (Include Zip Code) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS: Commander ATTN: Joseph P. Toy, Code 603A, Bldg. 2917 NAVSURFWARCENDIV, 300 Highway 361 Crane, IN 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT. Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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INCH-POUND

MIL-M-28787/414
AMENDMENT 1
3 August 1994

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-1397 TYPE E SERIAL INTERFACE ADAPTER MODULE,

KEY CODE ATC

This amendment forms a part of Military Specification MIL-M-28787/414, dated 23 March 1994, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4 AND 5

TABLE I: Delete and substitute new section of table I as printed on pages 3 and 4 of this amendment.

PAGE 34

TABLE III, Test conditions B7 and B8, Max limit: Delete "1" (two places) and substitute "3" (two places).

TABLE III, NOTES, Continuity test tolerance table: Delete and substitute the following:

Condition	Value	Accuracy	Test condition
Continuity	NA	± 0.05 Ohm	B1-B6
Measurement	NA	± 0.15 Ohm	B7,B8

AMSC N/A
FSC 5963
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PAGE 61

TABLE XII, Test condition M1, Min limit: Delete "2.3" and substitute "3.4".

TABLE XII, Test condition M1, Max limit: Delete "4.3" and substitute "6.3".

TABLE XII, Test condition M2, Min limit: Delete "4.5" and substitute "3.4".

TABLE XII, Test condition M2, Max limit: Delete "8.3" and substitute "6.3".

TABLE XII, NOTES, Power supply filter capacitance test tolerance table, Accuracy column, Capacitance measurement: Delete "± 100 nF" for test condition M1 and "± 190 nF" for test condition M2 and substitute "± 145 nF" (two places).

PAGE 62

FIGURE 10: Delete and substitute new figure 10 as printed on page 5 of this amendment.

Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0297)

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TABLE I Module electrical requirements

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	S	1	INF	1	INF	1	INF	MOhm
A2 A50	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	X	1	INF	1	INF	1	INF	MOhm
B1 B6	Continuity	III	X	0	1	0	1	0	1	Ohm
B7 B8	Continuity	III	X	0	3	0	3	0	3	Ohm
C	Power supply stress	IV	S	----------Module shall pass all remaining tests ----------						
D1	Ibd	V	S	-0.02	1 02	-0 02	1 02	-0 02	1 02	mA
D2-D6	Ibd	V	S	0 0	0 1	0 0	0 1	0 0	0 1	mA
D7	Ibd	V	S	-0 58	1 14	-0 58	1 14	-0 58	1 14	mA
D8	Ibd	V	S	0 0	0 1	0 0	0 1	0 0	0 1	mA
D9	Ibd	V	S	-0 02	0 22	-0 02	0 22	-0 02	0 22	mA
D10 D11	Ibd	V	S	-0 02	1 02	-0 02	1 02	-0 02	1 02	mA
D12	Ibd	V	S	0 0	0 1	0 0	0 1	0 0	0 1	mA
D13	Ibd	V	S	-0 02	1 02	-0 02	1 02	-0 02	1 02	mA
D14	Ibd	V	S	0 0	0 1	0 0	0 1	0 0	0 1	mA
E1 E12	Ios	VI	S	-100 0	40 0	-100 0	-40 0	100 0	-40 0	mA
E13	Ios	VI	S	-85 0	-15 0	-85 0	-15 0	-85 0	-15 0	mA
E14	Ios	VI	S	-100 0	-40 0	-100 0	-40 0	-100 0	-40 0	mA
E15	Ios	VI	S	-85.0	-15 0	-85 0	-15.0	-85.0	-15 0	mA
E16	Ios	VI	S	-100 0	-40.0	-100 0	-40 0	-100.0	-40 0	mA
F1	Iih	VII	X	-3 04	-0.5	-3 04	-0 5	-3 04	-0 5	mA
F2 F6	Iih	VII	X	-0 4	0 02	-0 4	0 02	-0 4	0 02	mA
F7	Iih	VII	X	-6 64	-0 35	-6 64	-0 35	-6.64	-0 35	mA
F8	Iih	VII	X	-0 4	0 02	-0 4	0 02	-0 4	0 02	mA
F9	Iih	VII	X	-1 64	-0 51	-1 64	-0 51	-1 64	-0 51	mA
F10,F11	Iih	VII	X	-3 04	-0 5	-3 04	-0 5	-3 04	-0 5	mA
F12	Iih	VII	X	-0 4	0 02	-0 4	0 02	-0 4	0 02	mA
F13	Iih	VII	X	-3 04	-0 5	3 04	-0 5	-3 04	-0 5	mA

- See footnotes at end of table

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TABLE I Module electrical requirements - Continued

Test cond	Parameter 5/	Ref - test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F14	Iih	VII	X	-0 4	0 02	-0 4	0 02	-0 4	0 02	mA
G1	III	VIII	X	-3 04	-0 5	-3 04	-0 5	-3 04	-0 5	mA
G2 G6	III	VIII	X	-0 4	0 02	-0 4	0 02	-0 4	0 02	mA
G7	III	VIII	X	-6 64	-0.35	-6 64	-0 35	-6 64	-0 35	mA
G8	III	VIII	X	-0 4	0 02	-0 4	0 02	-0 4	0 02	mA
G9	III	VIII	X	-1 64	-0 51	-1 64	-0 51	-1 64	-0 51	mA
G10 G11	III	VIII	X	-3 04	-0 5	-3 04	-0 5	-3 04	-0 5	mA
G12	III	VIII	X	-0 4	0 02	-0 4	0 02	-0 4	0 02	mA
G13	III	VIII	X	-3 04	-0 5	-3 04	-0 5	-3 04	-0 5	mA
G14	III	VIII	X	-0 4	0 02	-0 4	0 02	-0.4	0 02	mA
H1 H16	Voh	IX	X	2 4	4 6	2 4	4 5	2 4	4 5	V
J1 J16	Vol	X	X	0 0	0 5	0 0	0 5	0 0	0 5	V
L1 L8	Vic	XI	S	-1 5	0 0	-1 5	0 0	-1 5	0 0	V
M1	PSFC	XII	S	3 4	6 3	3 4	6 3	3 4	6 3	µF
M2	PSFC	XII	S	3 4	6 3	3 4	6 3	3 4	6 3	µF
M3	PSFC	XII	S	2 4	4 6	2 4	4 6	2 4	4 6	µF
N1	Icc	XIII	X	520	1230	520	1230	520	1230	mA
N2	Icc	XIII	X	-249	-100	-249	-100	-249	-100	mA
P1,P2	Functional	XIV	X	See table XIV						
Q1,Q2	Dynamic	XV	S	See table XV						
Q3,Q4	Dynamic	XV	Q	See table XV						
R1 R6	Cin	XVII	Q	2	75	2	75	2	75	pF
R7	Cin	XVII	Q	10	235	10	235	10	235	pF
R8	Cin	XVII	Q	2	100	2	100	2	100	pF
R9	Cin	XVII	Q	2	125	2	125	2	125	pF
R10	Cin	XVII	Q	4	75	4	75	4	75	pF
R11	Cin	XVII	Q	4	100	4	100	4	100	pF

See footnotes at end of table

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